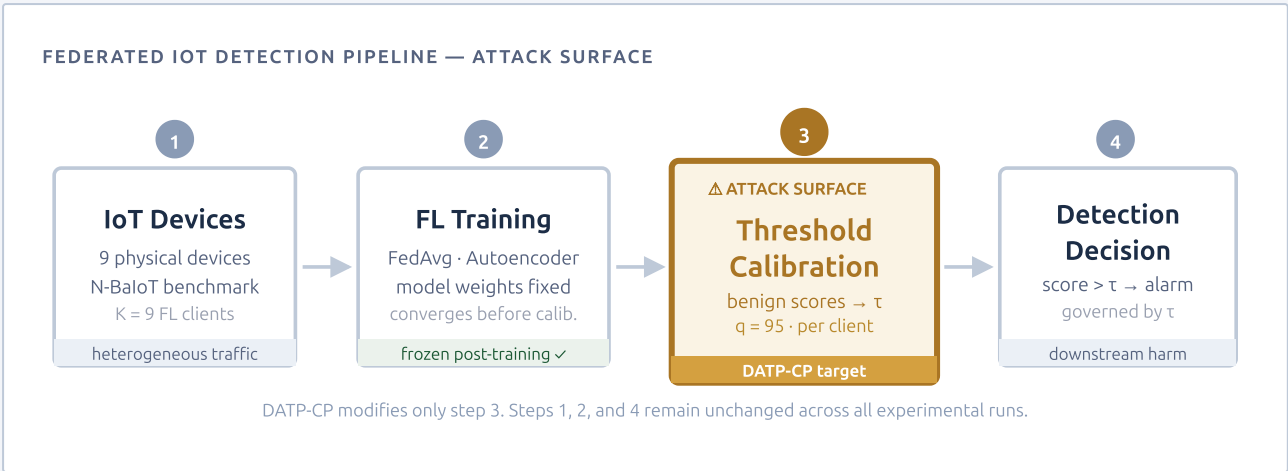


DATP-CP

Research Dossier

Scientific and Technical Scope for Calibration-Channel Poisoning
in Federated IoT Anomaly Detection



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RESEARCH SCOPE IN BRIEF

DATP-CP studies whether the post-training threshold-calibration stage in federated IoT anomaly detection is vulnerable to score-level poisoning. The attack modifies only benign calibration scores; training data, model weights, gradients, aggregation, test scores, and test labels remain unchanged throughout. The study compares **GLOBAL_THRESHOLD**, **LOCAL_THRESHOLD**, and **CLUSTER_THRESHOLD** under single-client compromise using the N-BaloT benchmark (9 physical IoT devices). No training poisoning, model poisoning, raw-traffic attack generation, privacy guarantee, or deployment-readiness claim is made.

Field	Value
Acronym	DATP-CP
Full title	Calibration-Channel Poisoning in Federated IoT Anomaly Detection
Document type	Research Dossier — Scientific and Technical Scoping Document
Author	Salaheddine Naslouby
Laboratory	LAMSAD Laboratory, ENSA Hassan First University, Berrechid
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Thesis line	Collaborative Malware Detection with Federated Learning in IoT Networks
Research branch	Doctoral Thesis — Branch 2 (Security)
Main dataset	N-BaloT (9 physical IoT devices as FL clients)
Object of study	Post-training threshold-calibration channel in federated anomaly detection
Version	v1.0
Date	July 2026
Status	Official scientific scoping document — not a paper, not a presentation deck

Document Purpose

This document defines the scope, objectives, methodology, deliverables, validation criteria, and boundaries of the DATP-CP research branch. It is intended as an official scientific and technical framing document, not as a full paper or presentation deck.

DATP-CP is the security branch of the DATP doctoral research line. It asks whether the post-training threshold-calibration stage in federated IoT anomaly detection is vulnerable to adversarial manipulation, under the specific constraint that training data, model weights, gradients, aggregation, and test evaluation are held unchanged.

All claims in this document are bounded to the nine-device N-BaloT setting with a fixed FedAvg autoencoder under three threshold policies: **GLOBAL_THRESHOLD**, **LOCAL_THRESHOLD**, and **CLUSTER_THRESHOLD**.

Document Organization

§	Section	Contents
1	Project Identity	Formal identity table and document purpose statement (this page)
2	Context and Rationale	DATP research line, three-branch structure, calibration coverage gap
3	Problem Statement	Problem, research problem, central research question, scientific boundary
4	Objectives	Scientific, technical, and doctoral objectives; pre-specified claim gates
5	Scope and Exclusions	Complete included and excluded elements with scope rationale
6	Methodology and Validation	Dataset, model, calibration, attack mechanism, and validation criteria
7	Deliverables, Risks, Governance	Deliverables, work packages, risks and limitations, closing statement

The DATP Research Line

DATP (*Device-Aware Threshold Personalization*) established that the threshold-calibration stage — not model architecture or training strategy — is the decisive variable governing per-client false-alarm burden in non-IID federated IoT anomaly detection. Using nine physical devices from the N-BalIoT benchmark as FL clients, DATP showed that a shared global threshold imposes severely unequal false-positive rates across devices ($CV(FPR) = 1.017$), while per-client calibration reduces this disparity substantially ($CV(FPR) = 0.299$), and that an unsupervised cluster policy recovers a significant fraction of the improvement without requiring device taxonomy labels.

DATP's controlled experimental design — fixing the FedAvg autoencoder and varying only the threshold policy — yields causally attributable results: the effect is due to calibration scope, not to training or architecture differences.

DATP-CP is the security branch of this research line. It takes the logical next step: if threshold calibration is the decisive stage for detection performance and fairness, is it also a viable attack surface? This question is precisely what DATP-CP is designed to answer, under a controlled score-level contamination model that preserves all of DATP's isolation properties.

Three Branches — Distinct Scientific Roles

Branch	DATP (Branch 1)	DATP-CP (Branch 2)	Journal Extension (Branch 3)
Scientific role	Threshold personalization and detection equity	Calibration-channel vulnerability analysis	Validation and generalization of DATP
Primary variable	Threshold policy scope	Calibration buffer contamination fraction f	Dataset and comparator breadth
Adversary	None — performance study	Gray-box, single compromised client	None — validation study
Primary claim	Calibration scope governs detection equity	Calibration channel is a viable attack surface	DATP effect survives generalization pressure

KEY PRINCIPLE

The three branches share a thesis line but have distinct scientific roles. DATP-CP is the adversarial robustness branch focused on the calibration stage. It does not absorb the journal extension's validation goals and does not duplicate DATP's personalization study. The journal extension — which targets external datasets, model-personalization comparators, and temporal recalibration — is explicitly out of scope for DATP-CP.

WHY CALIBRATION SPECIFICALLY?

Threshold calibration occurs after training convergence. Existing federated learning defenses — Byzantine-robust aggregation, gradient sanitization, model-integrity checks — have already exited by this stage. Each client autonomously collects its benign calibration window; the server cannot observe or sanitize it. This structural gap is the scientific motivation for DATP-CP.

Problem

Federated IoT anomaly detectors use benign calibration reconstruction scores to set per-client detection thresholds after training. This post-training calibration stage determines the operating point of the detector — it governs the balance between false alarms and missed detections for each client. It is structurally separate from training and aggregation, is executed autonomously by each client, and is never validated by a central authority.

Research Problem

CORE SECURITY QUESTION

Can a compromised client manipulate only its benign calibration buffer — without touching training data, labels, model weights, gradients, aggregation, test scores, or test labels — and still shift its detection threshold enough to produce material, interpretable downstream harm?

Central Research Question

DATP-CP CENTRAL RESEARCH QUESTION

Does poisoning only the benign threshold-calibration scores of eligible clients produce material, policy-dependent downstream harm under **GLOBAL_THRESHOLD** **LOCAL_THRESHOLD** and **CLUSTER_THRESHOLD** — and do these threshold policies exhibit different vulnerability profiles under the same calibration contamination?

Why This Matters

Security implications

- Thresholds control detection decisions — false negatives and false positives.
- Calibration is separate from training but equally security-critical.
- No existing FL defense covers the post-training calibration stage.
- Policy choice may determine victim sensitivity and collateral spillover.

Scientific implications

- DATP showed calibration scope governs performance equity.
- DATP-CP asks whether that same stage is adversarially exploitable.
- The causal isolation from training and aggregation is reproducible.
- Policy-differentiated results inform threshold policy selection under adversarial conditions.

Scientific boundary (non-negotiable): DATP-CP never claims training poisoning, model poisoning, aggregation poisoning, raw-traffic attack generation, privacy guarantees, universal FL robustness, or deployment readiness. These exclusions are permanent scope controls, not provisional limitations.

DATP-CP pursues three classes of objectives — scientific, technical, and doctoral — that jointly define what a successful research outcome requires and how it positions within the thesis.

4.1 Scientific Objectives	4.2 Technical Objectives	4.3 Doctoral Objectives
1. Isolate threshold calibration as an attack surface in federated IoT anomaly detection. 2. Measure material threshold movement under score-level benign calibration poisoning. 3. Compare vulnerability profiles across GLOBAL_THRESHOLD, LOCAL_THRESHOLD, and CLUSTER_THRESHOLD. 4. Characterize threshold-raising and threshold-lowering as separable failure modes. 5. Preserve causal isolation from training-phase and aggregation-phase poisoning throughout.	1. Validate or regenerate clean DATP-compatible artifacts for the N-BalIoT federation. 2. Implement controlled benign-score replacement under the REPLACE_FIXED_BUDGET injection rule. 3. Enforce valid source-objective combinations across all experimental conditions. 4. Produce paired clean-vs-poisoned comparisons across all threshold policies and poison fractions. 5. Generate reproducible figures, tables, manifests, and run logs for the reproducibility package.	1. Position DATP-CP as the security branch of the doctoral thesis, complementary to DATP and separate from the journal extension. 2. Complement DATP without duplicating its performance and personalization results. 3. Remain fully separate from the journal extension's validation and generalization goals. 4. Produce a defensible conference-style contribution bounded by pre-specified claim gates.

Pre-specified claim gates	Claim classes								
<table><tr><th>Gate</th><th>Pass condition</th></tr><tr><td>Gate 1</td><td>Seed-sign count $\geq 8/10$ AND victim-majority count $\geq 8/10$ for correctly-signed material $\Delta\tau$</td></tr><tr><td>Gate 2</td><td>Directional excess over matched RANDOM_BENIGN control at $\geq 8/10$ seeds</td></tr><tr><td>Gate 3</td><td>At least one downstream harm metric shows expected-sign support at $\geq 8/10$ seeds</td></tr></table>	Gate	Pass condition	Gate 1	Seed-sign count $\geq 8/10$ AND victim-majority count $\geq 8/10$ for correctly-signed material $\Delta\tau$	Gate 2	Directional excess over matched RANDOM_BENIGN control at $\geq 8/10$ seeds	Gate 3	At least one downstream harm metric shows expected-sign support at $\geq 8/10$ seeds	Depending on gate outcomes, the final claim is classified as: full vulnerability (all three gates pass), mechanism-only (Gate 1 only), calibration instability (Gate 2 fails), or null / conditional result. This pre-specification prevents post-hoc claim adjustment.
Gate	Pass condition								
Gate 1	Seed-sign count $\geq 8/10$ AND victim-majority count $\geq 8/10$ for correctly-signed material $\Delta\tau$								
Gate 2	Directional excess over matched RANDOM_BENIGN control at $\geq 8/10$ seeds								
Gate 3	At least one downstream harm metric shows expected-sign support at $\geq 8/10$ seeds								

The following tables define the complete scope of DATP-CP. Items in the left table are required experimental and methodological elements. Items in the right table are permanently excluded. The exclusions are intentional scope controls that preserve causal isolation — not provisional limitations.

Included in DATP-CP	Excluded from DATP-CP
✓ N-BaloT — main study dataset	✗ Training-data poisoning
✓ Physical IoT device as FL client	✗ Model poisoning
✓ FedAvg autoencoder — fixed trained model	✗ Aggregation poisoning
✓ Benign calibration-score poisoning	✗ Backdoor attacks
✓ Score-level proxy attack mechanism	✗ Evasion attacks
✓ GLOBAL_THRESHOLD policy	✗ Privacy mechanisms
✓ LOCAL_THRESHOLD policy	✗ Deployment benchmarking
✓ CLUSTER_THRESHOLD policy	✗ Raw-traffic attack generation
✓ q = 95 (p95 threshold calibration rule)	✗ Journal-extension datasets as main evidence
✓ Single-client compromise — main study	✗ Model-personalization comparators
✓ Paired clean-vs-poisoned evaluation	✗ Universal FL robustness claims
✓ RANDOM_BENIGN negative control	✗ Privacy guarantee claims

SCOPE NOTE

The exclusions are permanent scope controls, not provisional limitations. They preserve the causal isolation that makes the DATP-CP contribution defensible: any observed threshold shift is attributable to calibration contamination alone, since training, model, aggregation, and test evaluation are held unchanged across all experimental conditions. Relaxing these exclusions would invalidate the causal claim.

Experimental Setup

DATASET	
Dataset	N-BaloT
Clients (K)	9 physical IoT devices
Definition	One physical device = one FL client
Device types	5 (camera, doorbell, webcam, router, baby monitor)
Eligibility	n_cal ≥ 100 benign records
MODEL	
Architecture	Federated autoencoder
Aggregation	FedAvg
Model state	Fixed — frozen across clean-vs-poisoned runs
CALIBRATION	
Input	Benign reconstruction scores
Threshold rule	p95 — q = 95 (locked)
Policies	GLOBAL · LOCAL · CLUSTER

Attack Mechanism

Injection rule: REPLACE_FIXED_BUDGET

For victim client i and poison fraction f , select $m_i = \max(1, \text{round}(f \times n_i))$ positions in the victim's benign calibration array and replace them with values sampled with replacement from the declared victim-local reservoir. Calibration size remains constant — no records are added, only replaced.

Source strategies and objectives

Strategy	Reservoir	Objective
HIGH_SCORE_BENIGN	Top 10% victim-local benign	THRESHOLD_RAISE
LOW_SCORE_BENIGN	Bottom 10% victim-local benign	THRESHOLD_LOWER
RANDOM_BENIGN	Uniform victim-local sample	Negative control

Poison fractions: $f \in \{0.10, 0.20, 0.40\}$ · Seeds: 10 training seeds (NBAIOT_MAIN)

Validation Criteria

Criterion	Measurement	Role
Signed $\Delta\tau$	Correctly signed threshold shift (positive for raise, negative for lower)	Mechanism validation
Material $ \Delta\tau $	Threshold shift exceeds pre-specified materiality threshold	Gate 1
ΔTPR	Victim true-positive rate decrease under THRESHOLD_RAISE	Downstream harm (raising)
$\Delta\text{CV}(\text{FPR})$	FPR dispersion increase under THRESHOLD_LOWER	Downstream harm (lowering)
Policy differentiation	Distinct vulnerability profiles across all three threshold policies	Gate 2
Negative-control behavior	RANDOM_BENIGN yields near-null $\Delta\tau$ in same conditions	Confounding audit
AUROC invariance	AUROC unchanged across clean and poisoned runs	Sanity check only — calibration-channel isolation

Deliverables

#	Deliverable
D1	Protocol and roadmap (DATP_CP_Roadmap.md + this dossier)
D2	Implementation — attack engine, calibration injector, and pipeline
D3	Clean artifact audit — validated DATP-compatible artifacts
D4	Poisoning experiment outputs — NBAIoT_MAIN results across all conditions
D5	Figures and tables — paired clean-vs-poisoned visualizations
D6	Conference manuscript — DATP-CP paper draft
D7	Reproducibility package — code, manifests, seeds, and run logs
D8	Presentation dossier — visual research overview (separate document)

Work Packages

- WP1 Protocol lock and scope audit — finalize roadmap, claim gates, and scientific boundaries
- WP2 Clean artifact validation — validate or regenerate DATP-compatible N-BaloT artifacts
- WP3 Attack implementation and smoke tests — SYNTHETIC_SMOKE stage, unit validation
- WP4 N-BaloT main execution — NBAIoT_MAIN, 10 seeds, all policies, all fractions
- WP5 Analysis, figures, and tables — confirmatory inference, policy-differentiation analysis
- WP6 Manuscript preparation — conference-style paper, bounded to pre-specified claim class
- WP7 Final audit and packaging — FINAL_AUDIT stage, reproducibility package, forbidden-phrase check

Risks and Limitations

- R1 Score-level proxy: operates at reconstruction-score level — raw-traffic realizability is not claimed and is out of scope.
- R2 N-BaloT-only main evidence: all confirmatory claims are bounded to the nine-device N-BaloT setting.
- R3 Single-client main claim: multi-client extension (optional) is not required for the primary contribution.
- R4 No deployment benchmark: no claim is made about real-world FL system deployment or operational security.
- R5 AUROC invariance is a sanity check only, not a contribution claim — it confirms the calibration channel was correctly isolated.
- R6 With-replacement resampling at high fractions may duplicate score values — disclosed as a proxy limitation.

CLOSING STATEMENT

DATP-CP contributes a bounded, causally isolated, policy-differentiated analysis of a neglected post-training attack surface in federated IoT anomaly detection. By restricting the attack to the benign calibration channel and fixing training, model, aggregation, and test evaluation, the study achieves a causal attribution that broader observational comparisons cannot provide.